Vekto: Verifying Semantic Equivalence of SIMD Kernels Across Architectures

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1 Introduction

High-performance numerical software relies on kernels that exploit target-specific instructions and memory behavior. Developing these kernels requires substantial hardware expertise, and moving to a different architecture often entails another round of implementation and tuning. Recent advances in language models have made it possible to automate parts of this work, with agents using hardware descriptions and execution feedback to generate and optimize low-level code [11, 7].

We focus on SIMD kernels for CPU vector extensions. Their C intrinsic interfaces expose vector operations within ordinary C programs, allowing agents to modify instruction selection and loop organization directly. These rewrites must account for differences in the underlying execution models. Arm NEON uses fixed-width vectors, whereas RISC-V Vector (RVV) selects the number of active elements at runtime; a port must therefore account for different iteration sizes and tail accesses [2, 14]. Recent work has applied language models to this migration, including translation of NEON kernels to RVV followed by target-specific optimization [5, 6].

Correctness errors can arise in the transformations that improve performance: changing a loop's iteration size affects which elements are loaded, while replacing an intrinsic may change rounding or exceptional-value behavior. These errors can survive tests on ordinary inputs. Existing formal approaches already validate generated code through bounded symbolic execution or compiler-IR reasoning [5, 16, 9]. Establishing equivalence over the full calling domain requires reasoning about every admitted loop iteration and memory access, under a model that retains the relevant ISA differences. Even a one-element maximum can disagree across ISAs if one instruction propagates a NaN and the other selects the numerical operand (Section 2). Our objective is to verify the optimized implementation under an explicit calling contract, while preserving the code whose performance motivated the optimization.

This objective requires a semantic model for both the vector operations and the surrounding program. Machine-code and compiler-IR verification provide established ways to express such models [3, 15, 9, 1]. For intrinsic kernels, the C source already exposes the loops, pointer operations, and calls that must be related. We therefore give the supported intrinsics executable C interpretations and verify the resulting C programs. This keeps the proof obligations close to the source and allows intrinsic definitions and supporting lemmas to be reused across kernels. The interpretations retain architecture-specific behavior; equivalence is established between the complete kernels under their calling contracts.

We present Vekto, a framework for semantic equivalence checking of SIMD kernels using reusable C semantic shims. Its design targets NEON, RVV, AVX, and Hexagon HVX. [**TODO:** Report the supported intrinsic subsets.] A common specification determines the required outputs for the logical inputs, while each implementation has its own memory representation and loop invariants. Proving both implementations against this specification relates their results without requiring their loops to advance in lockstep. QCP generates verification conditions for the annotated C programs, which are discharged through automation and Rocq/Coq proofs [17]. These obligations cover the output relation and modeled memory accesses. The guarantee is partial correctness relative to the intrinsic models and calling contracts; Section 3 states its trust assumptions and scope.

VEKTO separates the implementation, specification, and proof so that agents can develop proof annotations and auxiliary lemmas without altering the executable algorithm. Experts review the calling contract and any changes suggested during verification. A feasible counterexample calls for repairing the target and checking the revised implementation. A counterexample excluded by the original caller's documented assumptions instead calls for making those assumptions explicit in the specification, with evidence that all legitimate calls remain covered. This separation makes the reason for each change reviewable and allows its performance cost to be measured independently of proof development.

The same verification method supports optimized-kernel validation, translation to another ISA, and cross-checking of independently developed implementations. For translation, an existing NEON or AVX kernel supplies a reference when a target library lacks the operation. For cross-checking, the question is whether implementations of the same operation agree under a shared contract, extending the comparison underlying differential testing to a program equivalence obligation [10, 8]. A concrete disagreement identifies behavior to examine against the library contract, helping distinguish a defect requiring repair from an intentional numerical difference.

Our evaluation will examine these uses through the 35 final optimized RVV kernels released by SALTyRN, ports of five AVX operations to RVV and HVX, and corresponding implementations in XNNPACK [4]. A composed SIMD program, with softmax as a candidate, will test whether the kernel contracts support reasoning across operation boundaries. [**TODO:** Report verification coverage and cost, justified repairs and their performance effects, cross-checking findings, and the composed-program result.] The main measurements are the optimized implementations that can be verified without executable changes, the effort required to extend the semantic models, and the performance retained after necessary repairs.

2 A motivating example: maximum across two vector ISAs

[**TODO:** Candidate example. The discrepancy and diagnostic measurements below have archived evidence. A complete proof of the repaired final optimized kernel is pending. This example does not establish a speedup over XNNPACK and therefore does not yet satisfy the intended performance narrative.]

Consider an elementwise maximum kernel that reads two binary32 arrays and writes one result per element. Figure 1 shows excerpts from a NEON reference and the final RVV implementation in SALTyRN's published optimized-kernel repository [4]. The NEON loop processes four elements at a time. Its tail loads another four elements but stores only the requested prefix. The RVV implementation uses the maximum active length for full blocks and selects a shorter length for the final remainder. These control flows require different arguments about the processed prefix and valid memory accesses. In particular, the NEON tail requires readable padding even when only one output element is requested.

The numerical operations also differ. For the Arm floating-point profile used in this example, `vmaxq_f32` propagates a NaN operand. RVV's `vfmax` returns the numerical operand when exactly one operand is NaN [2, 14]. Treating both instructions as a single mathematical maximum would erase the difference before any program proof began. The semantic shims instead retain separate lane functions. Figure 2 extracts these functions from the existing QCP shim: the Arm branch quiets signaling NaNs and preserves a NaN result, whereas the RVV branch selects the numerical operand or a canonical NaN. Vector shims apply the appropriate function to the active lanes. This code illustrates existing semantic definitions, not a newly completed hardware-refinement proof.

NEON: fixed-width main loop

```
for (; batch >= 4 * sizeof(float); batch -= 4 * sizeof(float)) {
    const float32x4_t va = vld1q_f32(input_a); input_a += 4;
    const float32x4_t vb = vld1q_f32(input_b); input_b += 4;

    float32x4_t vacc = vmaxq_f32(va, vb);

    vst1q_f32(output, vacc); output += 4;
  }
```

RVV: full blocks and a variable-length tail

```
  size_t n = batch / sizeof(float);
  const size_t vlmax = __riscv_vsetvlmax_e32m8();
  while (n >= vlmax) {
    vfloat32m8_t va = __riscv_vle32_v_f32m8(input_a, vlmax);
    vfloat32m8_t vb = __riscv_vle32_v_f32m8(input_b, vlmax);
    vfloat32m8_t vacc = __riscv_vfmax_vv_f32m8(va, vb, vlmax);
    __riscv_vse32_v_f32m8(output, vacc, vlmax);
    input_a += vlmax;
    input_b += vlmax;
    output += vlmax;
    n -= vlmax;
  }
  if (n > 0) {
    size_t vl = __riscv_vsetvl_e32m8(n);
    vfloat32m8_t va = __riscv_vle32_v_f32m8(input_a, vl);
    vfloat32m8_t vb = __riscv_vle32_v_f32m8(input_b, vl);
    vfloat32m8_t vacc = __riscv_vfmax_vv_f32m8(va, vb, vl);
    __riscv_vse32_v_f32m8(output, vacc, vl);
  }
```

Figure 1 Excerpts from the unchanged source implementations. Signatures, assertions, and the NEON tail are omitted here; the complete source files are retained in the draft package. The RVV comments are omitted without changing executable statements. `batch` counts bytes, and `n` counts float elements. The proof must cover the complete functions, including the omitted tail.

The disagreement has a one-element witness:

$$n = 1, \qquad A[0] = \texttt{0x7fc12345}\ \text{(quiet NaN)}, \qquad B[0] = \texttt{0x3f800000}\ (1.0).$$

With sufficient input padding and disjoint buffers, executing the compiled kernels under AArch64 and RVV instruction emulation produces NaN and 1.0, respectively. The RVV result is the same at the three tested hardware vector lengths, 128, 256, and 512 bits. This is a concrete execution discrepancy. It is not a difference in NaN payload or the sign of zero, both of which the output relation in Section 3 permits.

A diagnostic run of the local SALTyRN bounded checker accepted the same final RVV candidate at input length one and VLEN 512. Inspection showed that its NEON maximum model used the same floating-point maximum operation as the RVV model. In an isolated copy, adding the NEON NaN-propagation case changed the result to a counterexample without changing the kernel pair, input length, or comparison relation. Thus this particular missed discrepancy concerns the semantic model, not the input-length bound. The run used a locally adapted checker and is not a reconstruction of the exact verification run reported in SALTyRN's paper. It also does not demonstrate that QCP automatically synthesizes counterexamples.

To preserve the NEON output behavior, the candidate repair retains the RVV maximum and selects a canonical NaN wherever either operand is NaN:

```c
unsigned int qcp_fmax_arm_bits(unsigned int a,unsigned int b)
{
 if (qcp_fmax_snan_bits(a) == 1) return a | 4194304U;
 if (qcp_fmax_snan_bits(b) == 1) return b | 4194304U;
 if (qcp_fmax_nan_bits(a) == 1) return a;
 if (qcp_fmax_nan_bits(b) == 1) return b;
 return qcp_fmax_ordered_bits(a,b);
}

unsigned int qcp_fmax_rvv_bits(unsigned int a,unsigned int b)
{
 if (qcp_fmax_nan_bits(a) == 1) {
  if (qcp_fmax_nan_bits(b) == 1) return 2143289344U;
  return b;
 }
 if (qcp_fmax_nan_bits(b) == 1) return a;
 return qcp_fmax_ordered_bits(a,b);
}
```

Figure 2 Existing C lane interpretations, with their QCP annotations omitted for space. The helpers inspect binary32 bit patterns and compare non-NaN values. Decimal constants `4194304U` and `2143289344U` encode the quiet-NaN bit and a canonical quiet NaN. The original definitions and contracts accompany the excerpts. The modeled Arm profile disables flush-to-zero and default-NaN behavior; floating-point exception flags are outside the output observation.

```c
vbool4_t a_nan = __riscv_vmfne_vv_f32m8_b4(va, va, vl);
vbool4_t b_nan = __riscv_vmfne_vv_f32m8_b4(vb, vb, vl);
vbool4_t any_nan = __riscv_vmor_mm_b4(a_nan, b_nan, vl);
vacc = __riscv_vfmerge_vfm_f32m8(
    vacc, __builtin_nanf(""), any_nan, vl);
```

The repair is inserted after maximum in both the full-block and tail paths, using `vlmax` in the former. It leaves the NEON implementation, RVV loop organization, and register grouping unchanged. Regression executions agree under the intended output relation, including mixed exceptional and finite values. [**TODO:** Complete the final optimized kernel's shim bindings, loop and memory obligations, and all dependent proof checks for the same profile as the performance experiment. Historical proofs of an intermediate RVV implementation do not discharge these obligations.]

The measured geometric-mean speeds relative to XNNPACK u8v are 0.982× for the published optimized candidate and 0.826× for the repair (Figure 3). The repair increases the geometric mean of the per-size cycle ratios by 18.8%. These measurements do not support a claim that this candidate retains a performance advantage over XNNPACK. They instead expose the cost of preserving the selected reference behavior. Whether that behavior is required depends on the contract: XNNPACK's historical min/max discussion explicitly declines a NaN-handling guarantee [18]. Consequently this witness establishes failure of NEON-preserving equivalence, without by itself establishing an XNNPACK API defect.

3 Contracts, observations, and proof obligations

3.1 The relation being checked

Let x be a logical input, including array contents, dimensions, and operation parameters. A domain predicate $D(x)$ describes legitimate calls. For implementation K_i, a precondition $P_i(x, s_i)$ relates x to the initial C state s_i, including its memory representation. A machine profile p_i records the ABI, relevant floating-point configuration, and supported vector

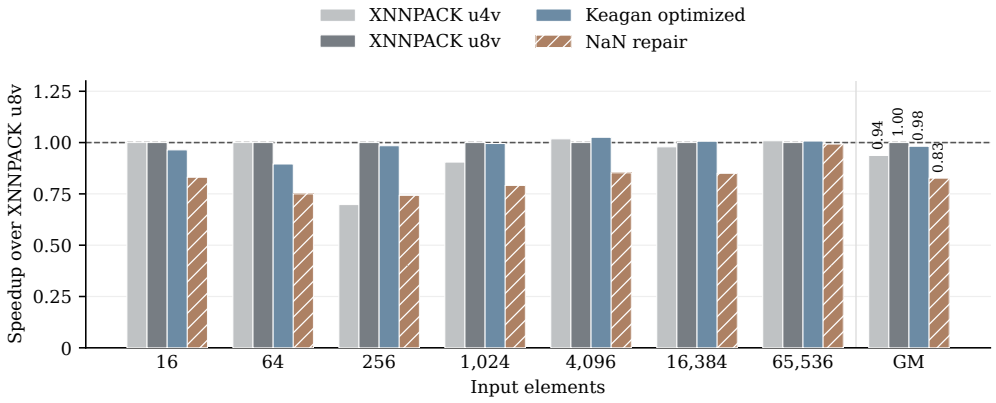


Figure 3 Preliminary local Saturn RTL measurements on finite inputs, reproduced from the archived experiment. Speed is XNNPACK RVV u8v cycles divided by the measured implementation's cycles. Each size uses one warm-up and the median of three timed calls within one simulation run; no confidence interval is claimed. Hardware VLEN is 512 bits and datapath width is 256 bits. The published optimized and XNNPACK implementations do not preserve the NEON NaN behavior; the repair adds that behavior. These are measurements, not proof results or a precise reproduction of the SALTyRN paper's configuration.

configuration. The two profiles need not use the same vector width or state representation.

For integer results we require exact equality. For the floating-point outputs considered here, define

$$u \approx_{\mathrm{fp}} v \quad \Longleftrightarrow \quad u =_{\mathrm{num}} v \ \vee \ (\mathsf{isNaN}(u) \wedge \mathsf{isNaN}(v)). \tag{1}$$

Numeric equality identifies positive and negative zero. This relation also identifies NaN payloads, but permits no tolerance between distinct finite numbers. We lift it pointwise to output arrays, requiring their shapes to agree. A different observation must be stated explicitly for a kernel whose caller inspects bits, indices, or additional state.

Write $\langle K_i, s_i \rangle \Downarrow_{p_i} t_i$ for a terminating modeled execution. The partial-correctness comparison is

$$\begin{aligned} &D(x) \wedge P_A(x, s_A) \wedge P_B(x, s_B) \ \wedge \\ &\langle K_A, s_A \rangle \Downarrow_{p_A} t_A \ \wedge \langle K_B, s_B \rangle \Downarrow_{p_B} t_B \quad \Longrightarrow \quad \mathsf{obs}(t_A) \approx \mathsf{obs}(t_B). \end{aligned} \tag{2}$$

Here obs extracts the specified output and return values. Memory safety and preservation of memory outside the permitted write footprint are separate obligations of the implementation contracts. They do not follow from Equation 2 alone. The current QCP proof packages inspected for this draft explicitly exclude a termination theorem. Accordingly, Equation 2 does not assert that either implementation returns, or that they agree on divergence. **[TODO:** If the final contribution includes total correctness, add and check termination arguments for the admitted implementations; do not infer them from return VCs.]

3.2 Relating implementations through a common specification

Each implementation is checked against a postcondition that relates its output to a common relation $S(x, o)$. To compose the results into equivalence, the common relation must determine

observations up to the selected equivalence:

$$S(x, o_1) \wedge S(x, o_2) \Longrightarrow o_1 \approx o_2. \tag{3}$$

This condition prevents a weak common specification, such as a broad error interval, from being mistaken for an equivalence proof. For a reference-preserving floating-point optimization, S may encode the reference's rounding and accumulation order rather than ideal real arithmetic. The proof must establish the result for each complete implementation, including helpers and tails. A wrapper that assumes both kernel contracts only establishes the final composition conditional on those contracts.

The relation in Equation 1 is a final-output observation. It cannot be substituted for bit-level equality at every intermediate boundary: a subsequent reciprocal or sign-sensitive operation can distinguish positive from negative zero. A composed program therefore needs intermediate contracts strong enough for each consumer, or a proof that the consumer respects the chosen relation.

3.3 What QCP checks

QCP processes contracts, assertions, and loop invariants by symbolic execution in separation logic [17, 12]. In the inspected VADD package, the public contract supplies readable input arrays, additional padding for NEON tail loads, and a separate writable output array. It returns the original input resources and a fully initialized output array with specified values. At an indexed read, generated entailments expose the cell at the requested index from the array predicate and retain the remainder of the array. Other conditions establish expression ranges, satisfy callee preconditions, preserve loop invariants, and derive the postcondition on return. Memory-access validity is checked through these resources and entailments as well as symbolic-execution failures; it is not confined to obligations named `safety` [13].

The proof package checks generated `VC_Correct` interfaces for the kernel bodies, shims, helpers, and final comparator. Statements declared inside such an interface are obligations to implement, not evidence that the obligations have been proved. The implementation/specification binding and dependency checks are therefore part of accepting a result. The inspected package establishes modeled C-shim partial correctness and explicitly excludes hardware-to-shim correspondence and termination. [**TODO:** Freeze the release and complete the per-kernel coverage audit for the final experimental cohort.]

3.4 Review and trust boundaries

Reusing a semantic shim requires a matching operation, type, and machine profile. In particular, a proof at one fixed RVV capacity is not a proof for all hardware VLENs. A sound generalization must account for the allowed active-length choices and the behavior of inactive or masked lanes. Source preparation must also preserve executable statements and their interpretation. Compiler correctness and the correspondence from modeled intrinsic semantics to native execution remain outside the present proof chain. The QCP paper reports that its symbolic executor lacked a formal soundness proof at publication [17]; [**TODO:** audit the exact QCP release used in the final artifact and state its remaining trusted components.]

When strengthening a precondition from P to P', let R describe the original caller's legitimate states. Acceptance requires evidence for $R \Rightarrow P'$, not merely that P' excludes one counterexample. Likewise, changing a target creates a new candidate whose proof and timing evidence must be rebound to the changed source. Source edits, contract changes, and

Table 1 Planned studies. Subject counts specify the intended evaluation, not proof successes. Softmax is a candidate composed program, not a full inference application.

Study	Subjects	Planned scope	Completed proofs
Optimization	Final SALTyRN targets	35 pairs	[**TODO:** pending]
Translation	AVX operations to RVV/HVX	5 operations	[**TODO:** pending]
Cross-checking	Corresponding XNNPACK kernels	[**TODO:** select]	[**TODO:** pending]
Composition	A small SIMD program	[**TODO:** select]	[**TODO:** pending]

Table 2 Per-kernel reporting schema. Each row must identify exact source versions and the checked profile in the artifact. Times include failed attempts where applicable. No placeholder is a measurement.

Field	Required value
Proof outcome	Proved, concrete discrepancy, or unresolved; [**TODO:** counts]
Contract/profile	Admitted lengths, values, memory layout, VLEN and FP modes
Executable changes	Added/deleted lines by ISA, rationale, review decision
Verification cost	Wall time, generation/checking time, tokens, retries; [**TODO:** measure]
Reuse	Reused intrinsic lemmas and assumptions; [**TODO:** measure incremental cost]
Performance	Baseline/original/repaired cycles and speed ratios; [**TODO:** measure]
External optimization cost	Upstream records, or explicitly not reported

proof-only changes are recorded separately. A failed proof, a replayed counterexample, and a completed proof are distinct outcomes.

4 Evaluation design

[**TODO:** This section specifies planned experiments. It contains no completed cohort-level verification or performance results.]

The primary cohort is fixed to the 35 published final optimized RVV kernels at commit `80feb4564cb4` [4]. The study evaluates existing optimization outputs; it does not rerun the optimization search. For each subject, the matching NEON source, original optimized target, any repair, caller contract, and machine profile must be identified separately. Earlier proofs of intermediate targets are reusable development assets, not completed results for this cohort.

Measurements should distinguish proof construction from proof checking. Per-kernel records include elapsed time, QCP condition generation, proof checking, agent token usage, failed attempts, manual interventions, and implementation changes. Shared semantic-library construction is reported separately from incremental kernel work. Historical optimization time and tokens are reported only when traceable upstream records exist. The study compares original optimized and repaired candidates against the same handwritten RVV baseline, using identical input sizes, compiler settings, and target hardware configuration. An invalid candidate's timing may quantify repair cost, but is not counted as a valid speedup.

The extension study must establish that selected operations are absent from the chosen target-library versions rather than infer absence from filenames. Cross-checking must align logical operations, layouts, parameters, and numerical contracts before classifying a discrepancy. Existing XNNPACK reports supply potential historical regression subjects, including RVV rounding and square-root exceptional-value handling [19, 21]. They are known reports,

not new findings by Vekto. For softmax, the comparison must address reduction order, exponential approximations, intermediate-buffer contracts, and the final division. [**TODO:** Select a real source program and target baseline, and establish whether the desired exact observation is compatible with their algorithms.]

Three case studies are planned: a minimally repaired optimized kernel; a new port that requires additional intrinsic semantics; and a cross-checking discrepancy whose feasibility and contract implications can be traced to concrete code. [**TODO:** Choose completed cases with adequate provenance. The maximum example currently supports semantic diagnosis, but not the desired retained-speedup claim.]

5 Related work

[**TODO:** Focused initial comparison. Complete the literature coverage before making a novelty claim.]

Generation and validation of optimized kernels.

KernelBench, Autocomp, and IntrinTrans use executable feedback to evaluate or improve generated kernels [11, 7, 6]. SALTyRN also includes bounded symbolic comparison before and after optimization [5]. LLM-Vectorizer combines generation with Alive2 validation and describes loop-alignment techniques and restrictions for its experiments [16]. These systems motivate comparing the input domain, semantic model, and complete program obligations of each guarantee, rather than grouping all prior validation as testing. Vekto's proposed evaluation asks whether modular C proofs can cover the actual optimized kernels under explicit contracts.

Semantic models and verified transformations.

Sail and Islaris provide a path from authoritative ISA specifications to machine-code reasoning [3, 15]. Alive2 checks LLVM refinement with bounded reasoning about loops [9]. Rake checks equivalence of synthesized vector instruction sequences for Hexagon HVX [1]. These operate at different boundaries from Vekto's C intrinsic interface. The proposed method uses QCP's existing C verification infrastructure [17]; reusable cross-ISA semantic definitions, the alignment of kernel contracts, and evidence about optimized implementations must supply the contribution beyond that infrastructure.

Cross-checking implementations.

Differential testing uses disagreements among implementations to identify candidate defects [10]. SymDiff checks equivalence and exposes behavioral differences between imperative programs [8]. Our proposed XNNPACK study applies an explicit observation relation to corresponding ISA implementations, while retaining the distinction between disagreement and violation of a required contract. The distinction has practical consequences: a historical XNNPACK min/max policy permits unspecified NaN behavior [18], while a later change to the repository's separate YNNPACK subsystem adds NaN propagation for a reported JAX reduction problem [20]. That change is evidence of an integration concern, not proof that our tested XNNPACK microkernels violate their API.

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